

Author response to the minimum reporting checklist

Paper title: *(fill in)*

Version (date): *(fill in)*

Manuscript ID (optional): *(fill in)*

Corresponding author: *(fill in)*

Code or data release (optional): *(URL, DOI, or repository tag/commit)*

Instructions. For each checklist item ID, provide a one to three sentence response and point to the exact location where the information is reported (main text section, SI section, figure, table, or repository path/commit). When space is limited, keep a one line summary in the main text and place full details in the SI.

Required items (must report in main text)

ID	Checklist item	Response (one to three sentences)	Where reported
D1	Data	<i>(fill in; one to three sentences)</i>	<i>(e.g., Main text Sec. X; SI Sec. Sx)</i>
S1	Splits and leakage	<i>(fill in; one to three sentences)</i>	<i>(e.g., Main text Sec. X; SI Sec. Sx)</i>
M1	Model specification	<i>(fill in; one to three sentences)</i>	<i>(e.g., Main text Sec. X; SI Sec. Sx)</i>
T1	Training and inference	<i>(fill in; one to three sentences)</i>	<i>(e.g., Main text Sec. X; SI Sec. Sx)</i>
G1	Generation protocol	<i>(fill in; one to three sentences)</i>	<i>(e.g., Main text Sec. X; SI Sec. Sx)</i>
E1	Evaluation	<i>(fill in; one to three sentences)</i>	<i>(e.g., Main text Sec. X; SI Sec. Sx)</i>
B1	Baselines and ablations	<i>(fill in; one to three sentences)</i>	<i>(e.g., Main text Sec. X; SI Sec. Sx)</i>
F1	Feasibility (thermodynamic vs synthetic)	<i>(fill in; one to three sentences)</i>	<i>(e.g., Main text Sec. X; SI Sec. Sx)</i>
R1	Reproducibility	<i>(fill in; one to three sentences)</i>	<i>(e.g., Main text Sec. X; SI Sec. Sx)</i>

Optional items (recommended)

ID	Checklist item	Response (one to three sentences)	Where reported
CL1	Closed loop or active learning	<i>(fill in; one to three sentences)</i>	<i>(e.g., Main text Sec. X; SI Sec. Sx)</i>
C1	Compute footprint	<i>(fill in; one to three sentences)</i>	<i>(e.g., Main text Sec. X; SI Sec. Sx)</i>
GOV1	Governance and responsible use	<i>(fill in; one to three sentences)</i>	<i>(e.g., Main text Sec. X; SI Sec. Sx)</i>
K1	Key number summary	<i>(fill in; one to three sentences)</i>	<i>(e.g., Main text Sec. X; SI Sec. Sx)</i>